

Lesson 2 worksheet

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1. Given $a=(2,-1)$ and $b=(-4,5)$, find constants λ and μ such that $5a+\lambda(4,6)=\mu b$ [4 marks]

2. Find the vector with magnitude 39 in the same direction as $(12,-5)$ [2 marks]

3. Given that the x-coordinate of B is less than the x-coordinate of D, write EB and ED as column vectors. [2 marks]

4. Vector q : magnitude $12\sqrt{5}$, direction $(6,-3)$. Vector r : magnitude $15\sqrt{2}$, direction $(-5,5)$. Find unit vector in direction $q+r$. [6 marks]

5. The parallelogram OABC has $OA = a$ and $OC = c$. Point D lies on CB such that $CD:DB = 3:1$. Lines AB and OD meet at E where $OE = hOD$ and $BE = kAB$. Find DE in terms of a , c and h . [4 marks]

6. The diagram shows triangle OAC. Point B lies on AC such that $AB:BC = p:q$. $OA = a$, $OB = b$ and $OC = c$. Show that $b = (qa + pc)/(q + p)$. [5 marks]

7. The diagram shows a parallelogram OABC. The point D divides the line OC in the ratio 2:3. $OA = a$ and $OC = c$. The point P lies on AD such that $OP = \lambda OB$ and $AP = \mu AD$, where λ and μ are scalars. Find two expressions for OP, each in terms of a , c and a scalar, and hence show that P divides both DA and OB in the ratio $m:n$, where m and n are integers to be found. [7 marks]

Answer key

1. $\lambda = -15/22$, $\mu = -20/11$
2. (36, -15)
3. $EB = (-5, 3)$, $ED = (5, -3)$
4. $q = (24, -12)$, $r = (-15, 15)$, unit vector $= (1/\sqrt{90})(9, 3)$
5. $CD = 3a/4$, $OD = c + 3a/4$, $OE = h(c + 3a/4)$, $DE = h(c + 3a/4) - (c + 3a/4)$
6. Use section formula with ratio $p:q$, express b in terms of a and c
7. $OP = \lambda(a + c)$; $OP = a + \mu(-a + 2c/5)$; $\lambda = 2/7$, $\mu = 5/7$; ratio 2:5